

Current Treatment Strategies for Acromegaly: An Evidence-Based Review

DeepSeek + PubMed RAG Literature Synthesis Report

1. Research Question and Search Strategy

Research question: Current treatment strategies for acromegaly

DeepSeek-optimized PubMed query: acromegaly/therapy[MeSH] AND treatment strategies[Title/Abstract]

English semantic translation: Current treatment strategies for acromegaly

Candidate PubMed articles: 18

2. Executive Summary

- Acromegaly is a chronic disorder caused by a growth hormone (GH)-secreting pituitary adenoma, associated with significant morbidity and a two- to fourfold increased mortality risk if untreated PMID:16262568.
- Transsphenoidal surgery remains the first-line treatment, offering rapid biochemical remission and normalization of survival, though cure rates are limited in invasive macroadenomas [PMID:16262568, PMID:41550875].
- Pharmacotherapy, including somatostatin analogs (SSAs), the GH receptor antagonist pegvisomant, and dopamine agonists, plays a critical role in patients not cured by surgery, with SSAs being the largest contributor to direct medical costs [PMID:30122081, PMID:40025677].
- Radiotherapy, both conventional fractionated and stereotactic radiosurgery, is an effective secondary modality achieving remission in 50–60% of patients, though with a latency of several years PMID:20887122.
- Multimodal and individualized management by specialized multidisciplinary teams is essential to optimize long-term outcomes, with emerging evidence supporting preoperative medical therapy to improve surgical results [PMID:31100716, PMID:41550875].
- Despite therapeutic advances, up to 55% of patients remain uncontrolled in real-world settings, highlighting the need for new agents and strategies PMID:38651260.

3. Introduction and Disease Overview

Acromegaly is a rare, chronic systemic disorder resulting from excessive secretion of growth hormone (GH) and insulin-like growth factor-1 (IGF-1), most commonly due to a pituitary adenoma. Active disease is associated with a two- to fourfold increased mortality risk, primarily from cardiovascular complications PMID:16262568. The incidence and prevalence of acromegaly have increased over recent decades, likely due to improved diagnostic tools and earlier detection PMID:41961594. Effective management aims to normalize GH/IGF-1 levels, reduce tumor mass, and alleviate comorbidities, requiring a multimodal approach delivered by highly specialized multidisciplinary teams PMID:31100716.

Evidence links: PMID:16262568 PMID:41961594 PMID:31100716

4. Surgical Management

Transsphenoidal surgery is the treatment of choice for acromegaly due to its potential for rapid biochemical cure and normalization of long-term survival PMID:16262568. However, surgical remission rates remain suboptimal for invasive macroadenomas and giant adenomas, even with advancements in neurosurgical techniques PMID:41550875. In the best surgical hands, cure is achieved in only a proportion of patients, necessitating secondary treatment modalities PMID:16262568. Preoperative medical therapy with first-generation somatostatin receptor ligands (fg-SRLs) has been explored to improve surgical outcomes, though its clinical significance is still under investigation PMID:41550875.

Evidence links: PMID:16262568PMID:41550875

5. Pharmacotherapy

Medical therapy is invaluable for patients not cured by surgery or those with contraindications to surgery. Somatostatin analogs (SSAs) are the mainstay of pharmacotherapy and the largest contributor to direct medical costs in acromegaly management worldwide PMID:30122081. The GH receptor antagonist pegvisomant and dopamine agonists provide alternative or adjunctive options, particularly in patients with co-secreting prolactin adenomas [PMID:41912299, PMID:40025677]. Combined first-line medical treatment has shown effectiveness in acromegaly with prolactin cosecretion PMID:38771697. Despite the availability of these agents, up to 55% of patients remain uncontrolled in real-world data due to factors such as poor adherence, inadequate tolerability, therapeutic inertia, and high costs PMID:38651260. Drug treatment strategies also address secondary diabetes, a common comorbidity resulting from GH excess or its treatment PMID:32633582.

Evidence links:

PMID:30122081PMID:41912299PMID:40025677PMID:38771697PMID:38651260PMID:32633582

6. Radiation Therapy

Radiation therapy is an established secondary treatment for acromegaly. Conventional fractionated radiotherapy induces remission in 50–60% of patients but requires several years to achieve biochemical control PMID:20887122. Stereotactic radiosurgery offers a more targeted approach with potential for faster remission, though long-term comparative data remain limited PMID:20887122. Radiation is typically reserved for patients with residual or recurrent tumors after surgery and inadequate response to medical therapy.

Evidence links: PMID:20887122

7. Multimodal and Individualized Management

Modern management of acromegaly emphasizes an individualized, proactive approach from diagnosis to long-term follow-up, integrating surgery, pharmacotherapy, and radiotherapy as needed PMID:31100716. The time to biochemical remission is a critical predictor of overall outcomes, underscoring the importance of early and effective intervention PMID:39373288. Management of acromegalic arthropathy and other comorbidities requires a comprehensive, multidisciplinary strategy, even in patients with controlled GH/IGF-1 levels PMID:39485592.

Evidence links: PMID:31100716PMID:39373288PMID:39485592

8. Latest Clinical / Research Progress

- Preoperative medical therapy with first-generation somatostatin receptor ligands is being investigated to improve surgical remission rates in invasive macroadenomas PMID:41550875.
- Combined first-line medical treatment has demonstrated effectiveness in acromegaly patients with GH and prolactin co-secreting adenomas PMID:38771697.
- New drug delivery mechanisms and investigational agents are under development to address the 55% of patients who remain uncontrolled on current therapies [PMID:38651260, PMID:40025677].
- Improved epidemiologic data and earlier diagnosis have contributed to better management and reduced early mortality risk PMID:41961594.

9. Future Research Directions

- Development of novel pharmacotherapies, including new somatostatin receptor ligands, oral formulations, and targeted agents, to improve disease control and patient adherence [PMID:38651260, PMID:40025677].
- Further research into the role and timing of preoperative medical therapy to enhance surgical outcomes PMID:41550875.

- Optimization of individualized treatment algorithms based on tumor characteristics, biochemical profiles, and patient-specific factors PMID:31100716.
- Long-term studies on the cost-effectiveness of emerging therapies and their impact on morbidity and mortality PMID:30122081.

10. Limitations

- The evidence synthesis is based on a limited set of PubMed articles, which may not capture all relevant literature or recent clinical trials.
- Many studies are retrospective or observational, limiting the strength of causal inferences regarding treatment efficacy.
- Real-world data on disease control rates may be influenced by adherence and access issues not fully addressed in clinical trials PMID:38651260.

Appendix: PubMed Reference List

PMID	Year	Journal	Title
16262568		Expert Opin Emerg Drugs	Treatment strategies for acromegaly.
41912299		Vitam Horm	Co-secreting pituitary adenomas in acromegaly: Classification, diagnosis, and treatment strategies.
32633582		Expert Opin Pharmacother	Drug treatment strategies for secondary diabetes in patients with acromegaly.
38651260		Expert Opin Investig Drugs	Investigational drugs for the treatment of acromegaly: new agents to transform therapy.
40025677		Expert Opin Pharmacother	Promising therapies for the treatment of acromegaly.
20887122		Neurosurg Focus	Radiation treatment strategies for acromegaly.
30122081		Expert Rev Pharmacoecon Outcomes Res	Cost-effectiveness of somatostatin analogues in the treatment of acromegaly.
31100716		Eur J Endocrinol	MANAGEMENT OF ENDOCRINE DISEASE: Individualised management of acromegaly.
41550875		Front Endocrinol (Lausanne)	Preoperative medical therapy for acromegaly: current knowledge and clinical significance.
39331882		Medicine (Baltimore)	Secondary diabetes mellitus in acromegaly: Case report and literature review.
39485592		Pituitary	Approach to the patient with controlled acromegaly and acromegalic arthropathy: clinical diagnosis and management.
39373288		Clin Endocrinol (Oxf)	Remission of Acromegaly: The Sooner the Better.
38771697		Eur J Endocrinol	Effectiveness of combined first-line medical treatment in acromegaly with prolactin cosecretion.
41961594		J Clin Endocrinol Metab	Changing understanding of acromegaly epidemiology and early mortality risk.
19003539		Pituitary	Future treatment strategies of aggressive pituitary tumors.
22640971		Orphanet J Rare Dis	McCune-Albright syndrome and the extraskeletal manifestations of fibrous dysplasia.
40544415		Pituitary	Predictive factors for post-therapeutic biochemical discordance in acromegaly: a monocentric analysis.
22705998		Curr Opin Pediatr	Marfan syndrome: from gene to therapy.

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